C+C Programing

ASSIGNMENT

Programing fundamental

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Program :- BS Computer Science

Section E

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Q1 Write a C+C Program to find the fictorial of a number

```
# include <iostream>
Using namespace std;
int main ( ) {
  int n;
  Long Long fact = 1;
Cout << "Enter a non negative number;
Cin >> n;
if (n<0) {
Cout <<"factorial is not defined for negative
number;
 }
else {
 for (int i=1, i<=n;; i++) {
  fact = fact*i;
 }
 cout <<"factorial ='<< fact;
   }
 return 0
   }
```

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Q 2. Input three numbers and find the largest number

```
# include <iostream>
Using namespace std;
 int main ( ) {
  int a, b, c;
  cout << "Enter three numbers;
  Cin >> a >> b >> c;
 if (a>= & & a >= c) {
 Cout << "Largest is" << a;
    }
else if (b>a & & b >= c) {
 Cout << "Largest is" << b;
    }
   else {
 Cout << "Largest is" << c;
    }
  return 0;

  }
```

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Q3 Calculate the roots of a quadratic equation.

```
# include <iostream>
# include <maths>
 Using namespace std;
  int main () {
  double a, b, c, d, x1, x2;

     Cout << "Enter a, b and c: ";
     cin >> a >> b >> c;
     if (a == 0) {
    cout << "This is not a quatratic equation";
      }
     else
      d = b * b * , 4 a * c;
    if(d < 0){
    cout << There are no real root ";
      }
     else if (d == 0) {
     x1 = -b / 2 * a );
     cout << "Equal root = " << X 1;
      }
     else {
     x1 = (-b + Sqrt(d)) / (2 * a);
     x2 = -b - Sqrt (d)) / (2 * a;)
     cout << "x1 = " << x1 << endl;
     Cout << "x2 = << x2 << endl;
      }
      }
      return 0;
        }
```

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Q4. Convert a number from 1 to 10 into a Roman numeral using switch.

```
# include <iostream>
Using Namespace std;
 int main ( ) {
   int num;
cout<<"Enter a number from 1 to 10";
 cin >> num;
   Switch (num) {
   case 1: cout << "I"; break;
   case 2: cout << "II"; break;
   case 3: cout <<"III"; break;
   case4: cout <<"IV"; break;
   Case5: cout << "UI"; break;
   Case6: cout <<"VII"; break;
   Case7: cout << "VII"; break;
   case 8: cout << "VIII"; break;
   Case 9: cout<<"IX"; break;
   case 10: cout << "X; break;
    default: cout <<"Invalid number";
     }
     return 0;

      }
```

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Q5 Output / DRY-RUN Questions:.

Q5 : find the output:-

```
int x = 10, y = 20
if (x < y)
   x = x + 5
else
   y = y + 5
cout << x << " " << y;
```

Output :

15 20

Q6 find the output of the for loop.

```
for (int i = 2; i <= 10; i += 2) {
   cout << i << " ";
}
```

Output : 2 4 6 8 10

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Q7 find the output

```
int x = 3;
for (int i = 1; i <= 3; i++) {
  x = x + i;
  }
  cout << x;
```

output 30

Q8: find the sum

```
int sum = 0;
for(int i = 1; i<=5; i++){
  sum + i * 2;
  }
  cout << sum;
```

output 30

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Q9: find the output of the nested if-else statements:-

```
int n = 7;
if (n>5) {
 if (n>10) {
   cout <<"A";
    }
    else {
    cout <<"B";
   }
   }
   cout <<"C";
```

correct output: BC

Q10: find the output of the Switch statement.

```
int choice = 2;
Switch (choice) {
case 1: Cout <<"one";
case 2: Cout <<"Two";
case 3: Cout <<"Three"; break;
default: cout <<"other";
  }
```

output TwoThree

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Q11 find the output of the loop.

```
int x=1;
for(int i= 1; i<=4; i++) {
  x*=2;
  }
 cout << x;
```

output : 16

Q12 Display even number from 0 to 10

```
# include <iostream>
 Using namespace std;
 int main () {
 for (int n=0; n<=10; n+=2) {
  cout << n << " ";
 return 0;
  }
```

output : 0 2 4 6 8 10

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